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Assignment No: - 6

Problem Statement:
Implement sentiment analysis using Recurrent Neural Networks, specifically LSTM (Long Short-Term Memory) or GRU (Gated Recurrent Unit), to classify text data (e.g., movie reviews, tweets) into positive or negative sentiment.

Objective:

- Learn how LSTM/GRU networks are applied to sequential text problems.
- Properly preprocess textual datasets for deep-learning pipelines.
- Implement a sentiment classifier based on LSTM or GRU layers.
- Evaluate model performance using metrics like accuracy, precision, recall, and F1-score.
- Visualize training and validation curves to detect overfitting or underfitting.
- Study how hyperparameters (embedding dimension, hidden units, learning rate, batch size) affect results.

S/W Packages and H/W Apparatus Used:

- **Operating System:** Windows/Linux/MacOS
- **Kernel:** Python 3.x
- **Tools:** Jupyter Notebook, Anaconda, Google Colab
- **Hardware:** CPU with minimum 8GB RAM; GPU recommended (NVIDIA CUDA-enabled) for faster training
- **Libraries:** TensorFlow/Keras, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, NLTK/Spacy

Theory:

Sentiment Analysis: The task of determining the emotional tone of text (positive, negative, or neutral). Classical classifiers (Naive Bayes, SVM) often fail to capture order and context in sequences. Recurrent architectures such as LSTM and GRU keep internal state across timesteps, allowing them to model word order and long-range dependencies.

LSTM (Long Short-Term Memory):

  - Uses memory cells to store long-term dependencies.
  - Gates (input, forget, output) control the flow of information.
  - Helps prevent vanishing/exploding gradient problems in long sequences.
- **GRU (Gated Recurrent Unit):**
  - Combines the forget and input gates into an update gate.
  - Simpler than LSTM, faster to train.
  - Performs comparably in many NLP tasks while using fewer parameters.

Structure of LSTM/GRU Sentiment Model:

- **Input Layer:** Encoded text sequences (tokenized and padded).
- **Embedding Layer:** Converts words into dense vector representations (pre-trained embeddings like GloVe or Word2Vec can also be used).
- **RNN Layer:** LSTM or GRU units capture sequential dependencies. Can be stacked for better performance.
- **Dense Layer:** Fully connected layer with ReLU activation for non-linear transformations.
- **Output Layer:** Sigmoid activation (binary classification) or Softmax activation (multi-class classification).
- **Loss Function:** Binary crossentropy (for positive/negative) or categorical crossentropy (for multi-class).

Methodology / Algorithm:

1. **Data Acquisition:**
    - Collect a labeled dataset (e.g., IMDB movie reviews, Twitter sentiment dataset).
    - Ensure dataset has balanced classes to prevent bias.
2. **Data Preprocessing:**
    - Remove punctuation, special characters, URLs, and stopwords.
    - Convert text to lowercase for uniformity.
    - Tokenize text into words or subwords.
    - Convert tokens to integer sequences using Tokenizer (Keras).
    - Pad sequences to a fixed length using pad_sequences (Keras).
3. **Model Architecture:**
    - **Embedding Layer:** 100–300 dimensional word vectors.
    - **LSTM/GRU Layer:** 128–256 units. Can use return_sequences=True for stacked layers.
    - **Dropout Layer:** Prevent overfitting. Typical values: 0.2–0.5.
    - **Dense Layer:** ReLU activation.
    - **Output Layer:** Sigmoid (binary) / Softmax (multi-class).
4. **Model Compilation:**
    - Optimizer: Adam
    - Loss: Binary/Categorical Crossentropy
    - Metrics: Accuracy, Precision, Recall, F1-score
5. **Model Training:**
    - Train for 5–20 epochs depending on dataset size.
    - Use validation_split=0.1 or a separate validation set.
    - Monitor for overfitting using early stopping callbacks.
6. **Evaluation:**
    - Compute accuracy, precision, recall, and F1-score on test data.
    - Confusion matrix to visualize predictions.
7. **Visualization:**
    - Plot training and validation loss/accuracy curves.
    - Optionally, visualize word embeddings using t-SNE or PCA.
8. **Prediction:**
    - Preprocess new text input and predict sentiment using the trained model.

Advantages:

- Captures long-term dependencies in text.
- Handles vanishing gradient problem better than vanilla RNNs.
- Can work with short and long text sequences.
- Captures contextual meaning of words in sentences.
- Scalable to larger datasets and multiple sentiment classes.

Limitations:

- Computationally expensive, especially on large datasets.
- Requires large labeled datasets for good accuracy.
- Long sequences may still pose challenges.
- Sensitive to preprocessing choices.
- Difficult to interpret model decisions (black-box nature).

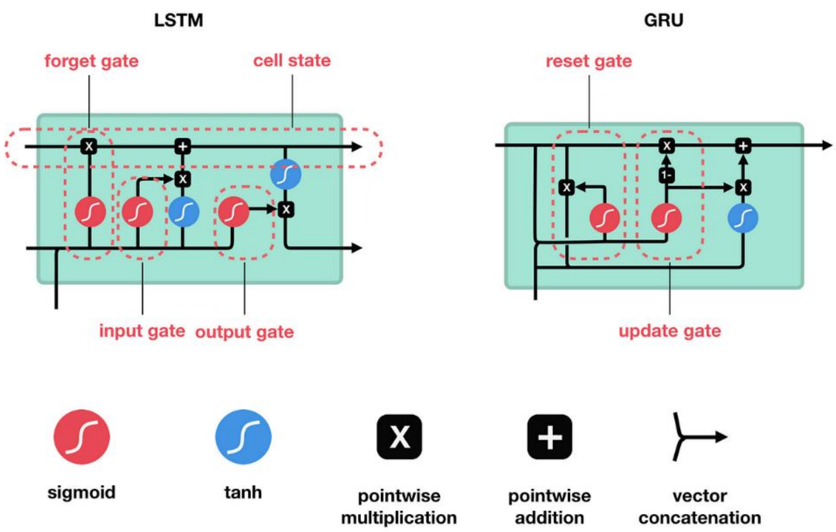
Applications:

- Customer feedback and review analysis (e.g., Amazon, TripAdvisor).
- Social media sentiment monitoring (Twitter, Facebook).
- Market research and brand perception analysis.
- Political opinion mining.
- Healthcare and patient feedback analysis.

Additional Notes / Enhancements:

- **Pre-trained Embeddings:** Using GloVe, FastText, or Word2Vec improves accuracy.
- **Bidirectional RNNs:** Capture context from both past and future words.
- **Attention Mechanism:** Enhances model by focusing on important words.
- **Hyperparameter Tuning:** Batch size, learning rate, number of units, embedding dimension can be tuned for best performance.
- **Data Augmentation:** Back-translation or synonym replacement can help with small datasets.

Diagram:



Conclusion:

Using LSTM or GRU networks for sentiment classification offers a powerful way to capture sequential dependencies and contextual meaning in text. Although these models demand computational resources, they typically outperform traditional methods when given adequate data and tuning. Incorporating pre-trained embeddings, bidirectionality, or attention can further improve results, making LSTM/GRU-based systems well suited for real-world sentiment analysis tasks across business, social media, healthcare, and political domains.